

Homework #1 — Probability and Counting (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Friday, 9/10 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as a .pdf format. Any submissions not in this format are subjected to point deductions.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

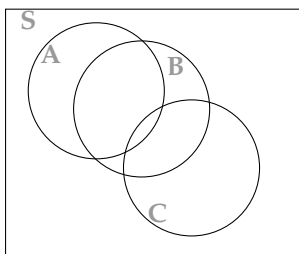
Background

The first homework will give you practice on using set notation, solving probabilities in a naive fashion, and identify/calculate probabilities of various subsets of events in a sample space.

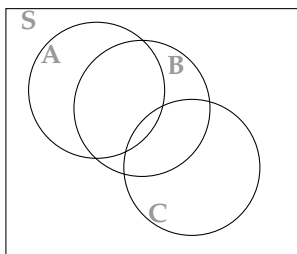
Part I: Venn Diagram (4 pts)

For each set operation below, you will shade in the corresponding area(s) of the Venn diagram.

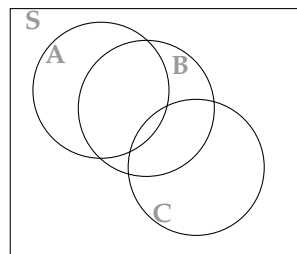
(a) $A^C \cap (B \cup C)$



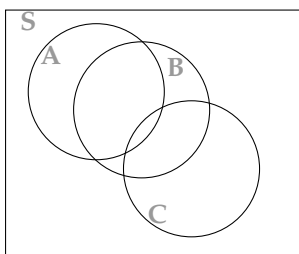
(b) $B \setminus (A \cap C)$



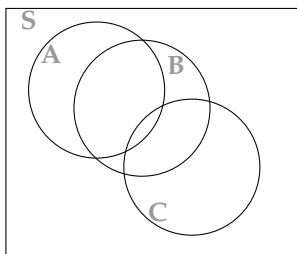
(c) $\emptyset^C$



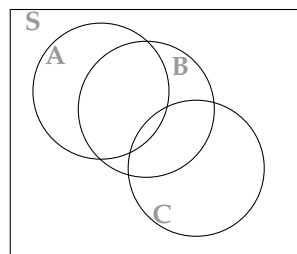
(d) $(A \cup B \cup C)^C$



(e) $(A \cap B) \cup (B \cap C) \cup (C \cap A)$



(f) $(A^C \cap B) \cup (C^C \cap A)$



Part II: 3 Rolled Dice (4 pts)

Suppose I roll 3 dice at the same time, where X is the outcome of the first die, Y is the outcome of second die, and Z is the outcome of the third die.

- (a) What is the sample space of this experiment? Consider the limitations of the dice when writing out the sets.
- (b) What is the cardinality of the sample space of this experiment?
- (c) Describe the following events in words and in terms of the outcomes (elements) of the sample space corresponding to the three rolls. (Ex: Let (x, y) be the outcome that the first die rolls $X = x$ and the second die rolls $Y = y$. Then the event that the two rolls sum to four is $\{X + Y = 4\} = \{(1, 3), (2, 2), (3, 1)\}$.
 - (i) $\{X + Y + Z = 4\}$
 - (ii) $\{X > Y + 3\}$
 - (iii) $\{X = Y - 1 = Z\}$
- (d) Find the probability of each of the aforementioned events occurring:
 - (i) $\{X + Y + Z = 4\}$
 - (ii) $\{X > Y + 4\}$
 - (iii) $\{X = Y - 1 = Z\}$

Part III: Football (4 pts)

Congratulations, you have just been hired as the head coach of the New York Jets. You decide to keep the play-calling very simple on the offense: all running plays.

Your running back has an equal chance of either gaining -2, 1, or 4 yards in a single play. You have 4 attempts to get 10 yards.

- (a) Identify the number of possible outcomes within four plays.
- (b) Identify all four-play outcomes that allow you to gain 10 yards. In other words, which permutations of four-play outcomes sums up to 10 or more?
- (c) What is the probability of getting at least 10 yards within 4 plays?

Part IV: Club (4 pts)

A census has been done on all 137 members of an arts & crafts club in the Hudson Valley. We have the following characteristics of these members:

- 58 of them are above the age of 60
- 100 of them enjoy drawing
- 47 of them are above the age of 60 and enjoy drawing

Find the following probabilities of...

- (a) being below the age of 60 in this club
- (b) being above the age of 60 and not enjoying drawing in this club
- (c) being below the age of 60 and enjoying drawing in this club
- (d) being above the age of 60 or enjoying drawing in this club

Part V: Textbook Problems (9 pts)

From *Introduction to Probability, Second Edition*:

- Chapter 1: Q34, Q55